

# CSE 370– Database Systems

## Makeup Assignment

Spring 2026

### Submission Instructions:

1. **Submit ONLY IF you DID NOT submit assignment 1.**
2. Write your name, id, section on top of the first page
3. Your answer should be handwritten, take pictures and create a single pdf.
4. **Submit the pdf through email ([najeefa.chy@bracu.ac.bd](mailto:najeefa.chy@bracu.ac.bd)) with the following subject:**  
CSE370 Makeup Assignment, and **write your name, ID and section** in the body of the email.
5. **Submission Deadline:** 16th May 2026 (Saturday), 11:59 pm (midnight)
6. **Marks will be deducted for not submitting Assignment 1. So you will not get full marks for this assignment.**
7. **NO LATE SUBMISSION WILL BE ACCEPTED, this is a makeup opportunity, if you do not submit, you will get 0. NO EXCUSES will be accepted.**

### Question 1

- A. **Construct** a B+ tree of order  $n = 3$  for the following search key values inserted in the given order: Apple, Orange, Grapes, Kiwi, Banana, Pineapple, Pear, Watermelon, Blueberry. Show the expansion of the tree for each insertion and the final tree. Values in nodes are sorted in alphabetical order.
- B. **Delete** the following from the B+ tree constructed in (A): Banana, Pear, Blueberry, Grape, Kiwi, Orange.

### Question 2

Sakib and Sarah are trying to reserve the same book- “DB Fundamentals” from an online library system. 5 copies of the book are remaining. T1 and T2 represent Sakib’s and Sarah’s transactions respectively. The concurrent transactions happen in the sequence given below:

1. T1: Sakib saw 5 copies left and reserved 1 book for himself.
2. T1: The number of copies is updated to 4, but not stored.
3. T2: Sarah also saw 5 copies left and reserved 2 copies for herself and her friend.
4. T2: The number of copies of the book is updated to 3 and is stored in the database.
5. T1: The number of copies is stored as 4 from Sakib’s transaction.

- a. Explain which ACID properties have been violated by the above transaction.
- b. State and explain all concurrency issues that have occurred due to the above transaction.
- c. State which isolation level(s) is/are guaranteed to prevent the concurrency issue(s) mentioned in (b).

### **Question 3**

Consider the following schema:

Organizers(ID, Name, Phone, Email)

Donors(ID, Name, Phone, Email)

Fundraiser(FID, Name, Description, Deadline, GoalAmount, OrgID) [FK: OrgID references ID of Organizers]

Donations(DonorID, FID, DonatedAmount, Date) [FK: DonorID references ID of Donors, FID references FID of Fundraiser]

Write Relational Algebra Expressions for the following questions:

- a. Retrieve the Donor Name, Donor Phone, DonatedAmount and Date for all donors who donated in the year 2026.
- b. Find the Name, Phone and email of all people who are both Donors and Organizers.
- c. Find the Name, Phone and email of all people who are Donors but not Organizers and the Donated Date and Amount of their donations.